Online Video Streaming System

PROBLEM STATEMENT

With the rapid growth of internet usage and digital media consumption, the demand for instant access to video content has significantly increased. Traditional methods of accessing videos often require full file downloads before viewing, which consumes considerable time, storage space, and bandwidth. This process is inefficient and inconvenient for users who expect immediate and uninterrupted playback.

Additionally, basic video systems do not provide real-time streaming with adaptive quality based on network conditions. Users frequently experience buffering issues, slow loading times, and inconsistent video resolution. There is also a lack of structured access control, subscription management, and secure digital payment integration for premium services.

From an administrative perspective, managing large volumes of video content, monitoring user access, tracking subscriptions, and maintaining system performance manually becomes complex and inefficient. Without an automated system, it is difficult to ensure smooth operation, accurate user management, and secure data handling.

Therefore, there is a need for a centralized Online Video Streaming System that enables real-time video playback without full download, supports adaptive streaming based on internet speed, provides secure authentication, manages subscriptions and payments digitally, and allows efficient administrative monitoring. The proposed system aims to improve user experience, reduce delays, enhance accessibility, and ensure reliable system performance.

ABSTRACT

The Online Video Streaming System is designed to provide users with instant access to video content over the internet without requiring full file downloads. Traditional video access methods consume significant storage space and cause delays before playback begins, often leading to poor user experience. Users frequently encounter buffering issues, slow loading times, and inconsistent video quality due to varying network conditions. The proposed system addresses these challenges by offering real-time video streaming with adaptive quality adjustment based on internet speed, ensuring smooth and uninterrupted playback. It includes secure user authentication, account management, subscription-based access, and digital payment integration for premium services. Administrators can efficiently monitor user activity, manage video content, and track subscription details through a centralized system. Designed for both web and mobile platforms, the system ensures scalability, reliability, and data security while optimizing bandwidth usage. Overall, the Online Video Streaming System provides a fast, secure, and user-friendly solution for modern digital video consumption.

INTRODUCTION

The rapid advancement of internet technology has significantly changed the way people consume digital media. Video streaming has become one of the most widely used online services for entertainment, education, and information. Traditional methods of accessing video content often require users to download the entire file before watching it, which consumes time,storage space, and high bandwidth. This process can lead to delays and inconvenience for users who expect instant access.

An Online Video Streaming System provides a solution by enabling real-time playback of videos without requiring full downloads. The system transmits video data in smaller segments, allowing playback to begin almost immediately. It also supports adaptive streaming, which automatically adjusts video quality based on network conditions to reduce buffering.

The system includes secure user authentication, subscription management, and digital payment integration to ensure controlled access and secure transactions. Administrators can monitor user activity, manage video data, and maintain system performance efficiently. Designed for both web and mobile platforms, the system ensures accessibility, scalability, and reliability. Overall, the Online Video Streaming System enhances user experience by providing a fast, secure, and efficient method of video delivery.

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

Online Video Streaming

Table of contents…….i

1. Introduction.....Error! Bookmark not defined.

1.1 Purpose.....Error! Bookmark not defined.

1.2 Document Conventions.....Error! Bookmark not defined. 1.3 Intended Audience and Reading Suggestions.....Error! Bookmark not defined.

1.4 Product Scope .....Error! Bookmark not defined.

1.5 References.....Error! Bookmark not defined.

2. Overall Description.....Error! Bookmark not defined.

2.1 Product Perspective.....Error! Bookmark not defined.

2.2 Product Functions .....Error! Bookmark not defined.

2.3 User Classes and Characteristics .....Error! Bookmark not defined.

2.4 Operating Environment.....Error! Bookmark not defined.

2.5 Design and Implementation Constraints.....Error! Bookmark not defined.

2.6 User Documentation .....Error! Bookmark not defined.

2.7 Assumptions and Dependencies .....Error! Bookmark not defined.

3. External Interface Requirements.....Error! Bookmark not defined.

3.1 User Interfaces.....Error! Bookmark not defined.

3.2 Hardware Interfaces.....Error! Bookmark not defined.

3.3 Software Interfaces .....Error! Bookmark not defined.

3.4 Communications Interfaces .....Error! Bookmark not defined.

4. System Features .....Error! Bookmark not defined.

4.1 System Features one by one.....Error! Bookmark not defined.

5. Other Non-functional Requirements.....5

5.1PerformanceRequirements.....5

5.2 Security Requirements.....Error! Bookmark not defined.

5.3 Safety Requirements .....Error! Bookmark not defined.

5.4 Software Quality Attributes.....Error! Bookmark not defined.

5.5 Business Rules.....Error! Bookmark not defined.

6. Other Requirements.....6

1. Introduction

The Online Video Streaming System is an application in which a user can log in and perform actions such as browsing videos, streaming content in real time, managing subscriptions, and viewing watch history. Through the system, users interact with a user-friendly interface that enables them to access video categories, search content, control playback settings, and check subscription status. The platform provides secure authentication, smooth video playback, and easy navigation, ensuring a convenient and efficient streaming experience for users across web and mobile devices.

1.1 Purpose

The purpose of the Online Video Streaming System is to provide a web and mobile-based platform that enables users to watch video content in real time without downloading the entire file. The system ensures smooth playback, secure user authentication, subscription management, and efficient administrative monitoring.

This document describes the functional and non-functional requirements of the system and serves as a reference for developers, testers, project managers, and stakeholders.

1.2 Definitions, Acronyms, and Abbreviations

Streaming – The process of playing video content in real time while it is being transmitted over the internet without downloading the complete file.

● **Adaptive Streaming** – A technique that automatically adjusts video quality based on the user's internet speed and device capability.

● **CDN (Content Delivery Network)** – A distributed network of servers that delivers video content efficiently to users based on their geographic location.

● **Buffering** – Temporary storage of video data before playback to ensure smooth streaming.

● **Bitrate** – The amount of data transmitted per second during video streaming.

● **Subscription Plan** – A paid access model that allows users to view premium content or features.

● **User Authentication** – The process of verifying user identity using login credentials such as email and password.

1.3 Product Scope

The Online Video Streaming System helps to provide users with easy access to video content and allows the management to maintain organized digital records instead of handling unstructured or manual processes. An automated video streaming system is far better than traditional download-based systems as it has the following advantages:

- Time Saving
- Increased Efficiency
- Accurate User and Subscription Management
- Decreases chances of buffering and delays
- Makes video access more organized and easier
- Allows proper handling of data instead of unstructured records

1.4 References

The references for the above software are as follows

i. www.google.co.in
ii. www.wikipedia.com

1.5 Overview

Section 1.0 discusses the purpose and scope of the software. Section 2.0 describes the overall functionalities and constraints of the software and user characteristics. Section 3.0 and 4.0 details all the requirements needed to design the software.

---

2. The Overall Description

2.1 Product Perspective

User Interfaces: The application has a simple and interactive graphical interface which makes it user-friendly. In this application, the user has several options to navigate through. The following screens will be provided:

(i) Login page is available for user authentication.
(ii) User has many options like Browse Videos, Search, Categories, Subscription Status, and Watch History.
(iii) Users can stream videos directly through the Video Player module.

2.2 Product Functions

Functions available to Users:

● Account Maintenance: Users are allowed to access their profile and can change the password for their account.

● Streaming and Watch History: Users can stream videos in real time and check their previously watched videos from the History tab.

● Browse and Search: Users can browse video categories and search for videos using the search bar on the Home page.

● Subscription & Payments: Users can check their subscription status, view plan details, and make payments through the Subscription tab.

● Video Controls: Users are provided with playback controls such as play, pause, seek, volume adjustment, and quality selection.

● Feedback/Rating: Users can give feedback or rate videos through the feedback section.

Functions available to Admin:

● Account Maintenance: Admin users are allowed to access their profile and change passwords.

● Content Management: Admin can manage the video database and remove inappropriate or outdated videos.

● Subscription Management: Admin can update subscription plans and monitor payment transactions completed till date.

● System Management: Admin can monitor user activity, generate reports, and update system details regarding premium features.

2.3 User Characteristics

User Classes:
● Guest Users: Able to browse available videos and stream limited content without registration.

● Registered Users: Able to access full streaming features, manage their subscription, view watch history, and provide feedback.
● Admin: Able to manage video records, monitor user activity, update subscription details, and maintain system operations.

User Characteristics:
a. Educational Level: Users should be comfortable with the English language.
b. Skills: Users should have basic knowledge and be comfortable using smartphones, web browsers, and general-purpose applications.

2.4 Operating Environment

The product will operate in a Windows environment and will also be compatible with modern operating systems such as macOS and Android/iOS platforms. Features will be compatible with major web browsers including Google Chrome, Mozilla Firefox, and Microsoft Edge. The application will also support access through smartphones and tablets. The only requirement is a stable and high-speed internet connection to ensure smooth video streaming without buffering interruptions.

2.5 Design and Implementation Constraints

● The user requires a computer or smartphone with an active internet connection.
● Proper authentication is required for user login and subscription access.
● Users should have a valid registered account with email ID or mobile number for verification.
● High bandwidth consumption and server load management are major technical challenges.
● Ensuring secure database connectivity and protection of user data is a critical constraint.

2.6 User Documentation

The product will include a user manual containing the system overview, configuration details, and technical specifications. It will explain how to register, log in, browse videos, manage subscriptions, and stream content. Basic troubleshooting steps and backup procedures will also be included for user and admin reference.

2.7 Assumptions and Dependencies

● It is assumed that users will have proper and stable internet access for smooth video streaming.
● It is assumed that users will use their accounts genuinely and follow subscription policies.
● It is assumed that users can properly operate smartphones or computers to access the application.

- The system depends on server availability and database connectivity for uninterrupted service.

---

3. System Features

- **User Login Screen:** Fields available on this screen are:
  - User Name / Email ID
  - Password
- **Video Browse Screen:** Fields available on this screen are:
  - Video Title
  - Category/Genre
  - Search Bar
  - Play Button
- **Subscription Plans:** Fields available on this screen are:
  - Plan Type (Monthly/Yearly)
  - Plan Price
  - Subscribe Option
  - Subscription Status
- **My Watchlist:** Fields available on this screen are:
  - Saved Video List
  - Remove from Watchlist Option
- **Streaming Player Interface:**
  - Video Player Window
  - Play/Pause Button
  - Volume Control
  - Full Screen Option
  - Video Quality Selection
- **Account Settings:**
  - Profile Details
  - Change Password
  - Subscription Details
- **Watch History:**
  - Recently Watched Videos
  - Date and Duration Watched
- **Feedback:** Fields available on this screen are:
  - Video Title
  - Rating Option
  - Comment Section

3.2 Hardware Interfaces

- Support for desktop computers, laptops, smartphones, tablets, and smart TVs for accessing and streaming video content.
- Compatibility with standard audio and video output devices such as speakers, headphones, and display screens for enhanced viewing experience.

3.3 Software Interfaces

● The application will operate on Windows, Android, and iOS-based operating systems.
● The software will transmit user data, streaming logs, and subscription details to a centralized database server via the internet.
● The admin will be allowed to modify video records, subscription plans, and user account details at any time.
● The system will integrate with a secure payment gateway for processing subscription payments.
● Notifications regarding subscription expiry and system updates will be managed through the application server Admin.

3.4 Communication Interfaces

The system needs to communicate with the central streaming server for each session to perform functions such as user authentication, video streaming, subscription verification, and data synchronization. The following requirements must be fulfilled:

- The user must connect to the Internet to access the Website/Application.
- A minimum internet speed of at least 1 Mbps is required for standard video streaming.
- A broadband or high-speed mobile data connection is recommended for uninterrupted HD streaming.
- The device must have an active internet service connection through a reliable Internet Service Provider.

---

4. External Interface Requirements

4.1.1 User Interface Requirements

- The interface should be simple and easy to navigate for users while browsing or streaming videos.
- Interfaces should be optimized with balanced graphics to ensure fast loading and smooth performance.
- Important buttons such as Play, Pause, Subscribe, and Logout should be clearly visible and accessible.
- The interface should display relevant error messages (e.g., "Invalid Login Credentials" or "Subscription Expired").
- The system should provide both web-based and mobile application interfaces accommodating all streaming functionalities.

4.1.2 Hardware Interface Requirements

- A screen resolution of at least 1024 × 768 is required for proper video viewing and interface display.
- Higher resolutions such as Full HD (1920 × 1080) or above will provide a better streaming experience.
- Minimum RAM of 2 GB is recommended for smooth application performance.
- Devices should support audio and video playback capabilities.

4.1.3 Software Interface Requirements

The software needs to interact with various systems to perform video streaming and account-related functions:

- Browsers supported: Google Chrome, Mozilla Firefox, Microsoft Edge, and Safari.
- Operating Systems supported: Windows, macOS, Linux, Android, and iOS.
- Browsers must be updated to the latest version for proper rendering and smooth video playback.
- A stable internet connection is required for streaming and interface loading.
- Proper login credentials are required for secure user access and subscription validation.

4.1.4 Communication Interface Requirements

The system must maintain continuous communication with the streaming server for session management, authentication, and video delivery.

- The user must connect to the Internet to access the platform.
- Proper configuration of network ports is required for secure client–server communication.
- A broadband or high-speed mobile data connection with an Internet Service Provider is recommended for smooth streaming.
- The network should support stable data transfer to avoid buffering or interruptions.
- Proper firewall configuration and security protocols must be enabled to ensure secure access to the platform.

---

5. Other Non-Functional Requirements

5.1 Performance Requirements

Some performance requirements identified for the Online Video Streaming Application are listed below:

- The website and mobile application must provide 24-hour service to ensure users can access and stream videos at any time.
- The system should process login, subscription validation, and video playback requests within a few seconds.
- The database shall be able to accommodate a minimum of 10,000 user records along with streaming history and subscription data.
- The software shall support multiple concurrent users streaming videos simultaneously without significant performance degradation.
- The system should minimize buffering time and maintain consistent video quality based on available bandwidth.

5.2 Security Requirements

 The database may fail due to cyber-attacks, malware, or system crashes; therefore, regular database backup and recovery mechanisms must be implemented.

 Security requirements include: Access control, Authentication and Authorization management, Disaster Recovery Planning and Data Backup for information systems, Network Security measures, Secure Coding and Application Security practices, Third-Party Vendor Security (for payment gateways), Encryption of user data, and Continuous Vulnerability Monitoring and Management.

5.3 **Safety Requirements**

The application will allow only valid registered users to access the system; security will be based upon individual User ID and Password authentication. Some of the factors identified to protect the software from accidental or malicious access, use, or modification are:

(i) Maintain specific logs or history data sets of user login activity, streaming sessions, and subscription transactions.

(ii) Assign certain functions to different modules, such as separating user streaming features from administrative account and subscription management functionalities.

The quality of the Online Video Streaming Application is maintained in such a way that it is user-friendly, performs efficiently, and satisfies the specifications given by the stakeholders. The software quality attributes are assumed as under:

● Usability: The software must be designed so that users can easily browse, search, and stream videos without confusion.

● Reliability: The system should be highly reliable and provide uninterrupted streaming with accurate subscription and account information.

● Portability: The software should run effectively on multiple platforms including Android, iOS, Windows, macOS, and various web browsers.

● Maintainability: The system should support easy updates, bug fixes, and feature enhancements in a cost-effective manner.

● Security: The system should resist malicious attacks, protect user data, and allow access only to legitimate users.

● Performance Efficiency: The system should respond quickly to search queries, login requests, and video playback actions.

● Flexibility: The system should be adaptable to future enhancements such as advanced recommendation systems or multi-language support.

● Scalability: The system should handle increased user traffic and streaming demand without degradation in performance.

5.5 Business Rules

The users holding an account in the Online Video Streaming Application must be registered with the platform.

● All users must associate a valid email ID or mobile number with their account, which will be used for OTP verification and password recovery.

● All users must have a valid User ID (Email/Mobile Number) and password to access the system.

● An account is considered valid only if the user can verify ownership of the registered email ID or mobile number.

● Users must accept the terms and conditions of the platform, including subscription policies and content usage rules.

● Users should not misuse the platform, such as sharing login credentials or attempting unauthorized access to premium content.

---

6. Other Requirements

● The system should support multiple video categories and genres for organized content browsing.

● Future integration with advanced technologies such as AI-based recommendation systems and smart TV platforms should be supported.

● Maintain logs for user activity, streaming sessions, and subscription payment transactions for monitoring and analysis purposes.